

Things You Wanted to Know!

Q1. What is FPGA? Please provide some notes and programs to explain it.

Irfan Khan

Through e-mail

A1. FPGA stands for field (in the field) programmable (re-configurable, change logic functions) gate array (reference to ASIC internal architecture). It is a semiconductor device containing programmable logic components called logic blocks, and programmable interconnects. It means that we can program it for almost any digital function.

The system works as follows: A computer is used to describe a logic function that is to be synthesised. Either a schematic or a text file describing the function will serve that purpose. Then, logic function is compiled, using the software provided by the FPGA vendor. This creates a binary file that can be downloaded into the FPGA. Next, FPGA is connected to a computer through a cable to download the binary file to the FPGA.

The programmed FPGA behaves according to logic function. However, do keep the following points in mind:

1. You can download FPGAs as many times as you want, with different functionalities every time if you want. If you make a mistake in your design, just fix your logic function, re-compile and re-download it. You do not need to change the PCB, soldering or component.

2. FPGA designs run much faster than designs with discrete components.

3. FPGAs lose their functionality when power goes away (like RAM in

a computer that loses its content). You have to download files again when power comes back, to restore functionality.

Learning material for the same is available on the Internet. EFY Tech Centers provide a nine-day hands-on short-term course on VLSI. The programming exercises are in Verilog and System Verilog languages, and tested on FPGA boards.

Q2. I want to know the points I need to keep in mind while selecting an RFID IC and microcontroller.

Anurag Bhist

Through e-mail

A2. Radio frequency identification (RFID) allows for wireless transfer of data between a small electronic transmitting tag and a reader without the necessity of line-of-sight. A feature of RFID is that the read operation may occur over long distances and multiple reads may occur. To choose the right RFID tag for your application, there are a number of considerations, including:

Frequency range. RFID products work on different frequencies, which are targeted for specific geographical regions, applications and performance requirements. Regulatory requirements associated for the region of operation are mandatory.

Memory size. This depends on the size of the database to be tagged.

Range performance. A tag's read-range performance is usually considered the primary gauge of its suitability for a particular application.

Environmental conditions. Where and how the tag or insert will be used plays a significant role in determining the right tag. Harsh environment, such as oil and gas exploration, construction and mining, will require rugged chips as compared to inventory and asset-tracking applications.

Cost of implementation and maintenance of the system. These are also to be considered while selecting the tag.

A microcontroller is selected primarily on the basis of application. There are a number of considerations, including:

1. General-purpose or specially-built for a specific application

2. Architectural features, which include:

- (i) 8-bit microcontroller for simple applications or 32-bit microcontroller for complex applications

- (ii) Number of ports to cater for input/output (I/O) requirements. If your requirement can be met with one input and one output pin, then you should opt for a 14-pin device instead of a 40-pin device which has a large number of I/O pins.

- (iii) Capacity of the program and data memory

- (iv) Availability of resident peripherals, such as timers, EEPROM, number of ADC channels, I2C, PWM, etc

- (v) Interrupt-handling capability

- (vi) Speed of operation

- (vii) Power requirement

3. Availability of development tools. Choose a microcontroller for which assemblers/compilers are available either as open source or are inexpensive.

4. Package. PDIP with leads suitable for normal PCB mounting or quad-flat package with no leads for SMD mounting

5. Easy availability in the market

To conclude, choose a microcontroller that meets the requirements of the application, and whose available features are used optimally.

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